Question (5): Maximize the Profit of a Software Company

The profit function is given as:

$$P(x_1, x_2) = x_1^3 + x_2^3 + 2x_1^2 + 4x_2^2 + 6,$$

where:

- x_1 = salary of employees,
- x_2 = general expenditure.

We need to:

1. Find the **Hessian matrix**.
2. Check whether the Hessian matrix is **positive definite** or **negative definite** at each stationary point.
3. Determine the values of x_1 and x_2 that **maximize the profit**.

Step 1: Stationary Points

Stationary points occur where the **partial derivatives** of the profit function with respect to x_1 and x_2 are zero. Let's calculate them.

First Partial Derivatives:

The derivative of $P(x_1, x_2)$ with respect to x_1 is:

$$\frac{\partial P}{\partial x_1} = 3x_1^2 + 4x_1.$$

The derivative of $P(x_1, x_2)$ with respect to x_2 is:

$$\frac{\partial P}{\partial x_2} = 3x_2^2 + 8x_2.$$

Set Partial Derivatives to Zero:

1. For x_1:

$$3x_1^2 + 4x_1 = 0.$$

Factorize:

$$x_1(3x_1 + 4) = 0.$$

Solutions:

$$x_1 = 0, \quad x_1 = -\frac{4}{3}.$$

2. For x_2:

$$3x_2^2 + 8x_2 = 0.$$

Factorize:

$$x_2(3x_2 + 8) = 0.$$

Solutions:

$$x_2 = 0, \quad x_2 = -\frac{8}{3}.$$

Stationary Points:

The stationary points are:

$$(x_1, x_2) = (0, 0),\ (0, -\frac{8}{3}),\ (-\frac{4}{3}, 0),\ (-\frac{4}{3}, -\frac{8}{3}).$$

Step 2: Hessian Matrix

The **Hessian matrix** is a square matrix of second-order partial derivatives. It helps determine the nature of the stationary points.

Second Partial Derivatives:

1. Compute the second derivatives with respect to x_1 and x_2:
   - $\frac{\partial^2 P}{\partial x_1^2} = 6x_1 + 4,$
   - $\frac{\partial^2 P}{\partial x_2^2} = 6x_2 + 8,$
   - $\frac{\partial^2 P}{\partial x_1 \partial x_2} = 0$ (no cross-terms).
2. Write the Hessian matrix:

$$H(x_1, x_2) = \begin{bmatrix} \frac{\partial^2 P}{\partial x_1^2} & \frac{\partial^2 P}{\partial x_1 \partial x_2} \\ \frac{\partial^2 P}{\partial x_1 \partial x_2} & \frac{\partial^2 P}{\partial x_2^2} \end{bmatrix}.$$

Substitute the derivatives:

$$H(x_1, x_2) = \begin{bmatrix} 6x_1 + 4 & 0 \\ 0 & 6x_2 + 8 \end{bmatrix}.$$

Step 3: Analyze the Nature of the Stationary Points

To determine whether a stationary point is a **maximum**, **minimum**, or **saddle point**, we check the **definiteness** of the Hessian matrix.

1. **Check Eigenvalues of the Hessian:**

- The eigenvalues of a diagonal matrix are simply the diagonal entries:
$$\lambda_1 = 6x_1 + 4, \quad \lambda_2 = 6x_2 + 8.$$

2. **Positive Definiteness:**
   - A matrix is positive definite if all eigenvalues are positive ($\lambda_1 > 0, \lambda_2 > 0$).
3. **Negative Definiteness:**
   - A matrix is negative definite if all eigenvalues are negative ($\lambda_1 < 0, \lambda_2 < 0$).

At Each Stationary Point:

1. **Point (0, 0):**
   - $\lambda_1 = 6(0) + 4 = 4$ (positive),
   - $\lambda_2 = 6(0) + 8 = 8$ (positive).

   Both eigenvalues are positive, so the Hessian is **positive definite**, and this is a **local minimum**.
2. **Point (0, -$\frac{8}{3}$):**
   - $\lambda_1 = 6(0) + 4 = 4$ (positive),
   - $\lambda_2 = 6(-\frac{8}{3}) + 8 = -8 + 8 = 0$.

   One eigenvalue is zero, so the Hessian is **not definite**, and this is a **saddle point**.
3. **Point (-$\frac{4}{3}$, 0):**
   - $\lambda_1 = 6(-\frac{4}{3}) + 4 = -8 + 4 = -4$ (negative),
   - $\lambda_2 = 6(0) + 8 = 8$ (positive).

   Mixed signs indicate the Hessian is **indefinite**, and this is a **saddle point**.
4. **Point (-$\frac{4}{3}$, -$\frac{8}{3}$):**

- $\lambda_1 = 6(-\frac{4}{3}) + 4 = -8 + 4 = -4$ (negative),
- $\lambda_2 = 6(-\frac{8}{3}) + 8 = -16 + 8 = -8$ (negative).

Both eigenvalues are negative, so the Hessian is **negative definite**, and this is a **local maximum**.

Step 4: Find the Global Maximum

From the analysis, the only **local maximum** occurs at:

$$(x_1, x_2) = (-\frac{4}{3}, -\frac{8}{3}).$$

Substitute these values into the profit function $P(x_1, x_2)$:

$$P(-\frac{4}{3}, -\frac{8}{3}) = (-\frac{4}{3})^3 + (-\frac{8}{3})^3 + 2(-\frac{4}{3})^2 + 4(-\frac{8}{3})^2 + 6.$$

1. Compute each term:
   - $(-\frac{4}{3})^3 = -\frac{64}{27}$,
   - $(-\frac{8}{3})^3 = -\frac{512}{27}$,
   - $2(-\frac{4}{3})^2 = 2 \cdot \frac{16}{9} = \frac{32}{9} = \frac{96}{27}$,
   - $4(-\frac{8}{3})^2 = 4 \cdot \frac{64}{9} = \frac{256}{9} = \frac{768}{27}$,
   - Constant term = 6.
2. Add them:

$$P = -\frac{64}{27} - \frac{512}{27} + \frac{96}{27} + \frac{768}{27} + 6.$$

Simplify:

$$P = \frac{288}{27} + 6 = \frac{288}{27} + \frac{162}{27} = \frac{450}{27} = 16.67.$$

Final Answer

The maximum profit is $\boxed{16.67}$, achieved at:

$$(x_1, x_2) = (-\frac{4}{3}, -\frac{8}{3}).$$